

Self-Learning: Red-Black Trees Data Structures

> **Definition:** A **Red-Black Tree** is a self-balancing Binary Search Tree where each node stores an extra bit representing **color** (Red or Black), satisfying specific balance constraints that ensure the tree height remains bounded by $2 \log_2(N + 1)$.

1. Detailed Technical Explanation

The 5 Red-Black Tree Properties:

1. **Node Color:** Every node is either **RED** or **BLACK**.
2. **Root Node:** The root node is always **BLACK**.

2. AVL Tree vs Red-Black Tree Comparison

Feature	AVL Tree	Red-Black Tree
Balance Factor	Must be -1, 0, or 1	Not applicable
Height Bound	$O(\log N)$	$O(2 \log N)$
Rotations	Requires rotations for rebalancing	Requires rotations for rebalancing

3. Insertion in Red-Black Tree

- When inserting a new key:
1. Insert the new node as a **RED** node.

4. Quick Recall Flow

RB-Tre